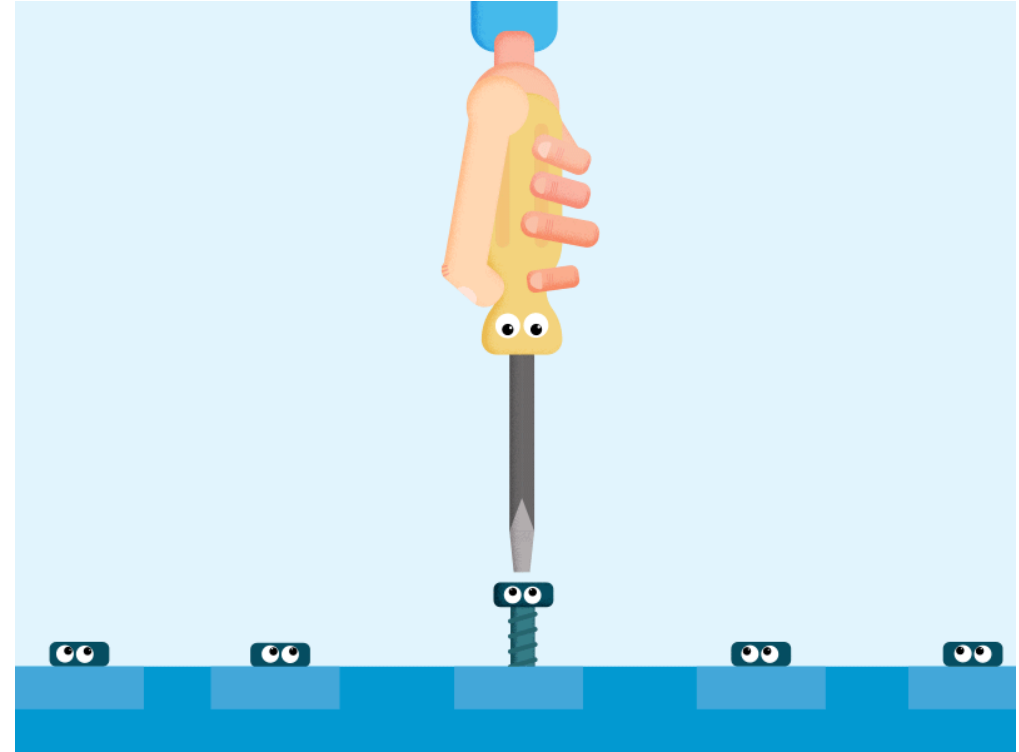


FITTING WORK

ME1031 – Engineering Skill Development

Department of Mechanical Engineering, Sri Lanka
Institute of Information Technology



GIF courtesy: <https://www.pinterest.com/pin/613826624192959813/>

Presented by Dr. L U Subasinghe | PhD (SG), BSc Eng Hons (SL), CEng.MIE(SL), AMIMechE
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LEARNING OUTCOMES

- At the end of this section, students should be able to
 - LO-6: Use of basic hand tools and machine tools in engineering workshop for machining processes and joining metals
 - **Fitting work**

OUTLINE

- Introduction
- General tools
- Fitting tools and operations

INTRODUCTION

INTRODUCTION

- Fitting shop is a workshop where various parts/components are assembled (fitted) together
 - Combination of machining and joining processes
 - Usually performed on a workbench
 - The person who assembles components together ➔ Fitter



A fitting workshop

GENERAL TOOLS

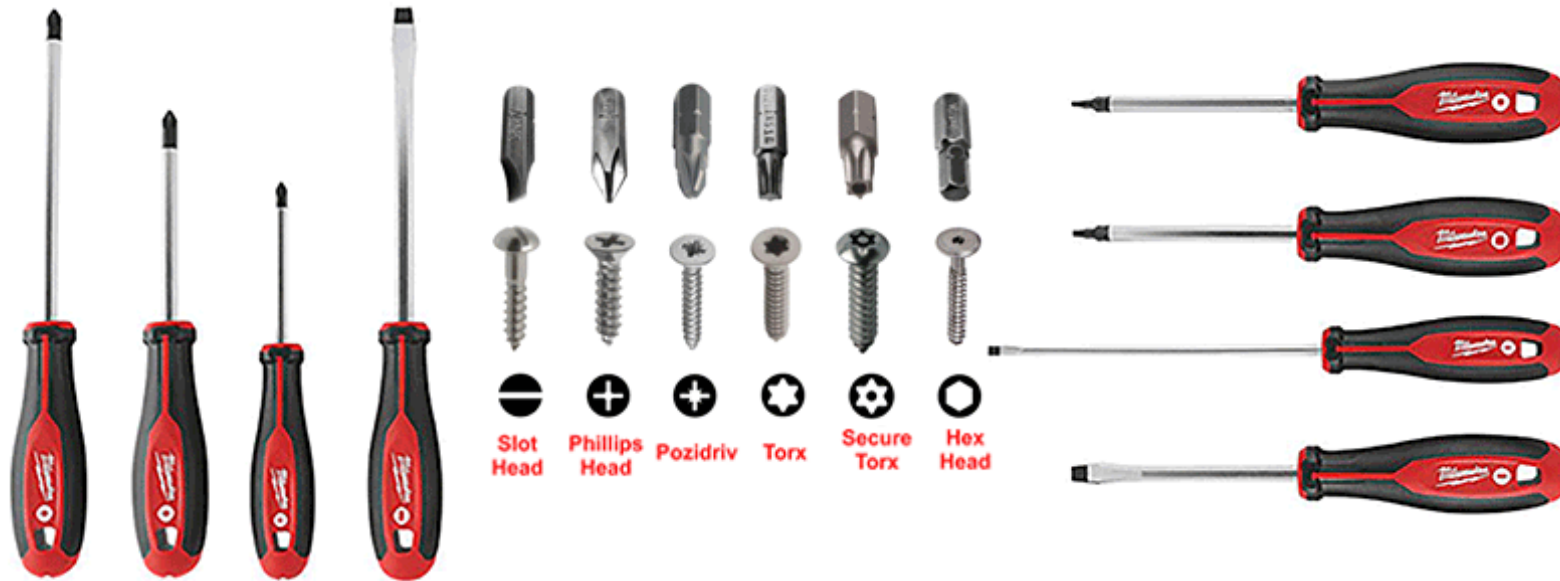
WHAT IS THIS TOOL?



A screwdriver

SCREWDRIVERS

Types Of



Screwdriver



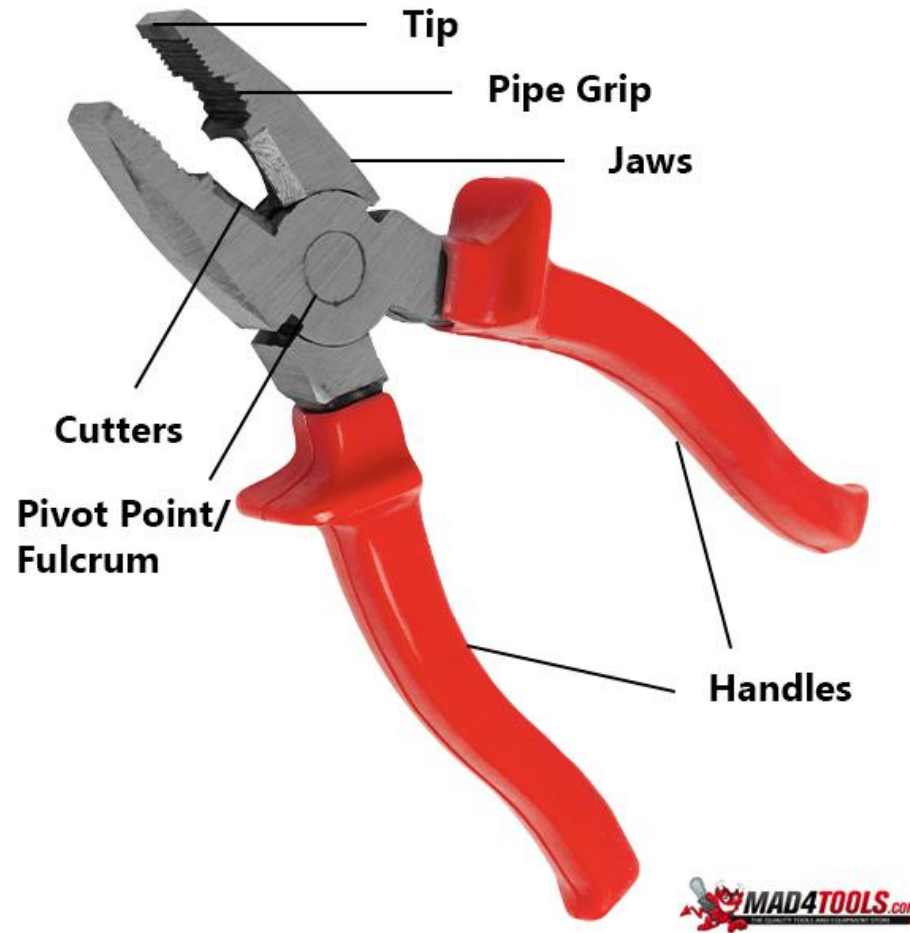
Allen keys (Hexagonal keys)



Hexagonal screws

<https://theunboxfactory.com/types-of-screwdrivers/>
https://en.wikipedia.org/wiki/Hex_key
https://www.fastflex.co.za/index.php?page=tape_types

WHAT IS THIS TOOL?

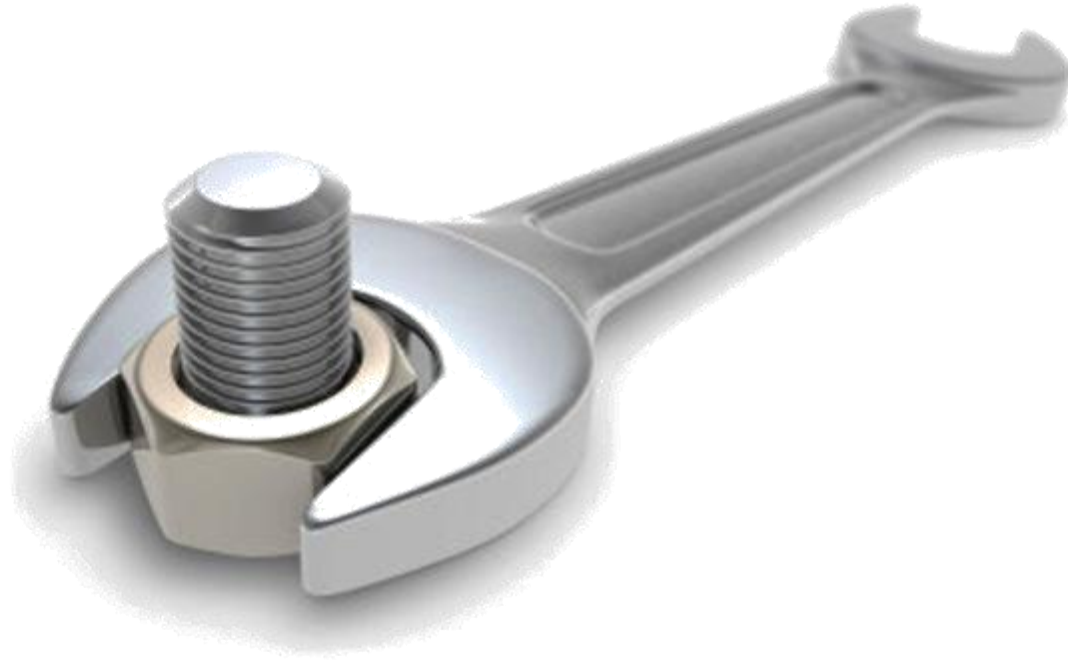


Pliers

PLIERS



WHAT IS THIS TOOL?



A spanner

SPANNERS



Open-ended spanner



Ring spanner



Combination spanner



Shifting spanner



Box spanner and tommy bar



Wheel brace



A ratchet, 4 hexagonal sockets and a U-joint

https://en.wikipedia.org/wiki/Socket_wrench
http://wiki.dtonline.org/index.php/Spanners_and_Wrenches
<https://en.wikipedia.org/wiki/Wrench>

DIFFERENT TYPES OF ADHESIVE TAPES

- Cellophane tape (Cellotape)
- Masking tape
- Duct tape
- Double-sided tape
- Insulation tape (Electrical tape)
- Kapton tape (Thermal insulation tape)



Cellophane tape



Masking tape



Kapton tape



Insulation tape



Duct tape

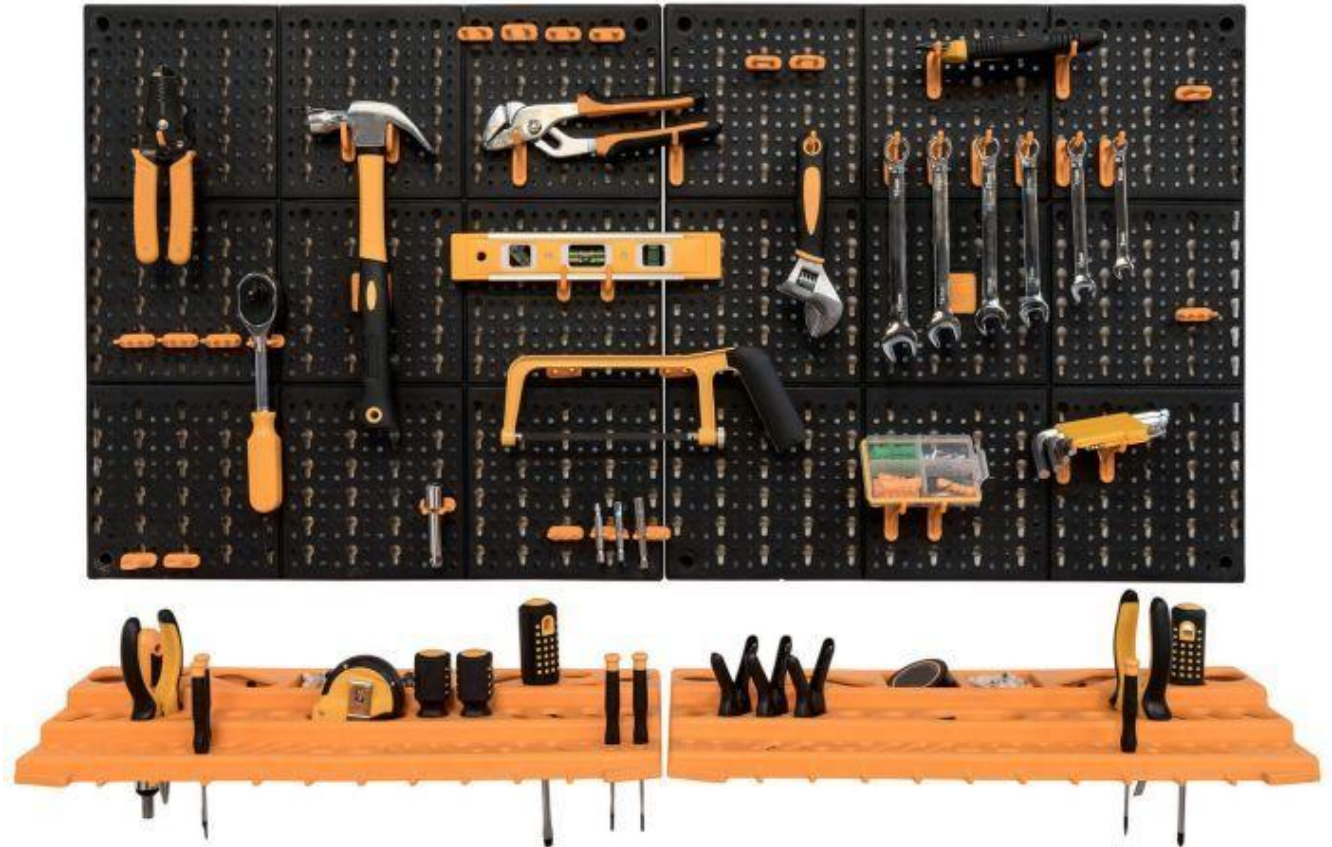


Double-sided tape

FITTING TOOLS AND OPERATIONS

FITTING TOOLS

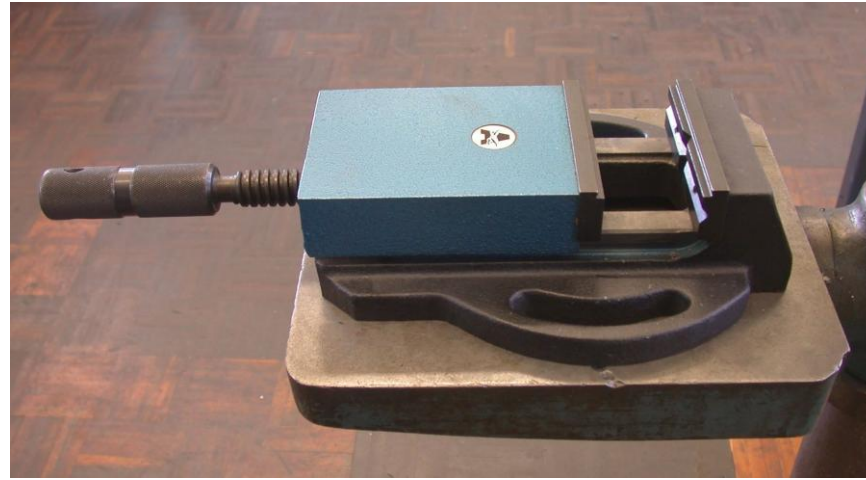
- Workholding tools
- Marking tools
- Measuring tools
- Cutting tools
- Finishing tools
- Striking tools



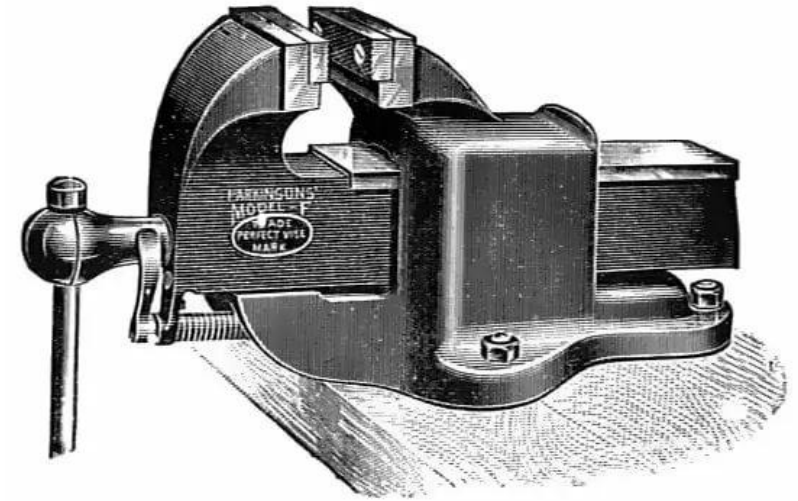
Fitting tools

WORKHOLDING TOOLS

- Workbench
- Bench Vice
- Pipe Vice
- Hand Vice
- Machine Vice
- V-block



Machine vice



Bench vice



Hand vice



Pipe vice



V block

MARKING TOOLS

- Scribe
- Dot Punch
- Center Punch

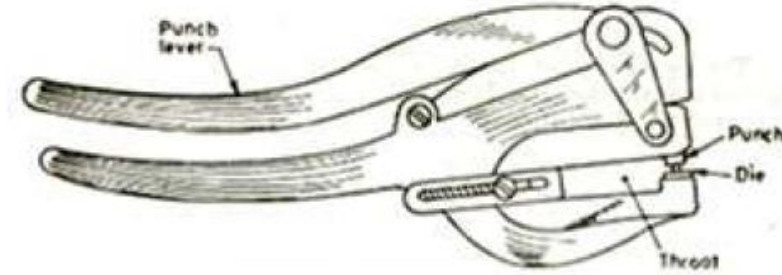
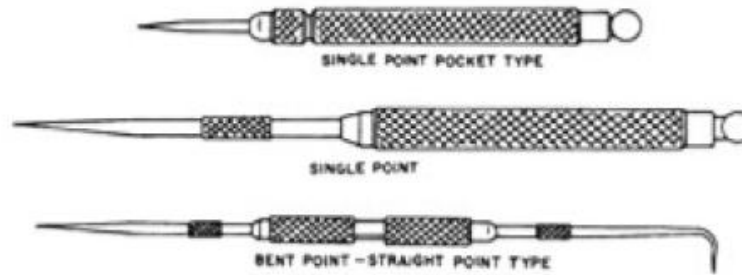
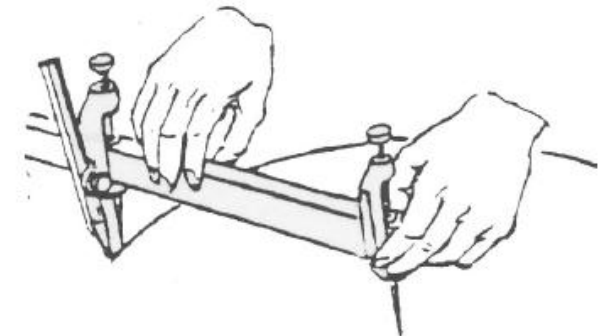
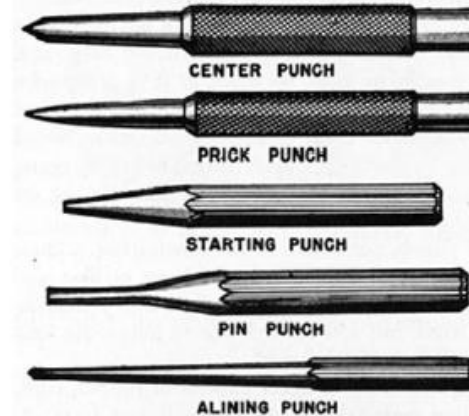
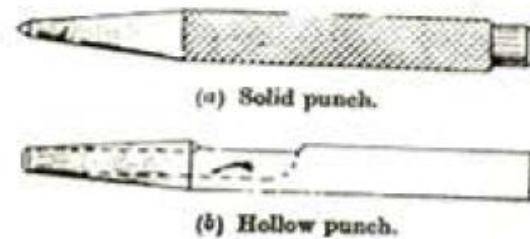
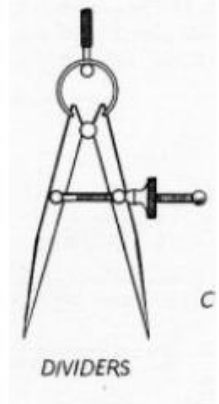


Fig. 15-8, Hand lever punch.



trammel points

MEASURING TOOLS

- Steel ruler
- Measuring Tape
- Surface Plate
- Divider
- Try square
- Inside and outside callipers
- Odd-leg caliper
- Vernier caliper
- Vernier height gauge
- Micrometre screw gauge



Surface plate



Divider



Try square



Odd-leg caliper



Inside caliper



Outside caliper

CUTTING TOOLS

- Hacksaw
- Chisel
- Drilling tool
- Reaming tool
- Tap and die



Taps



Dies



Tap with a "T" wrench (or tap handle)

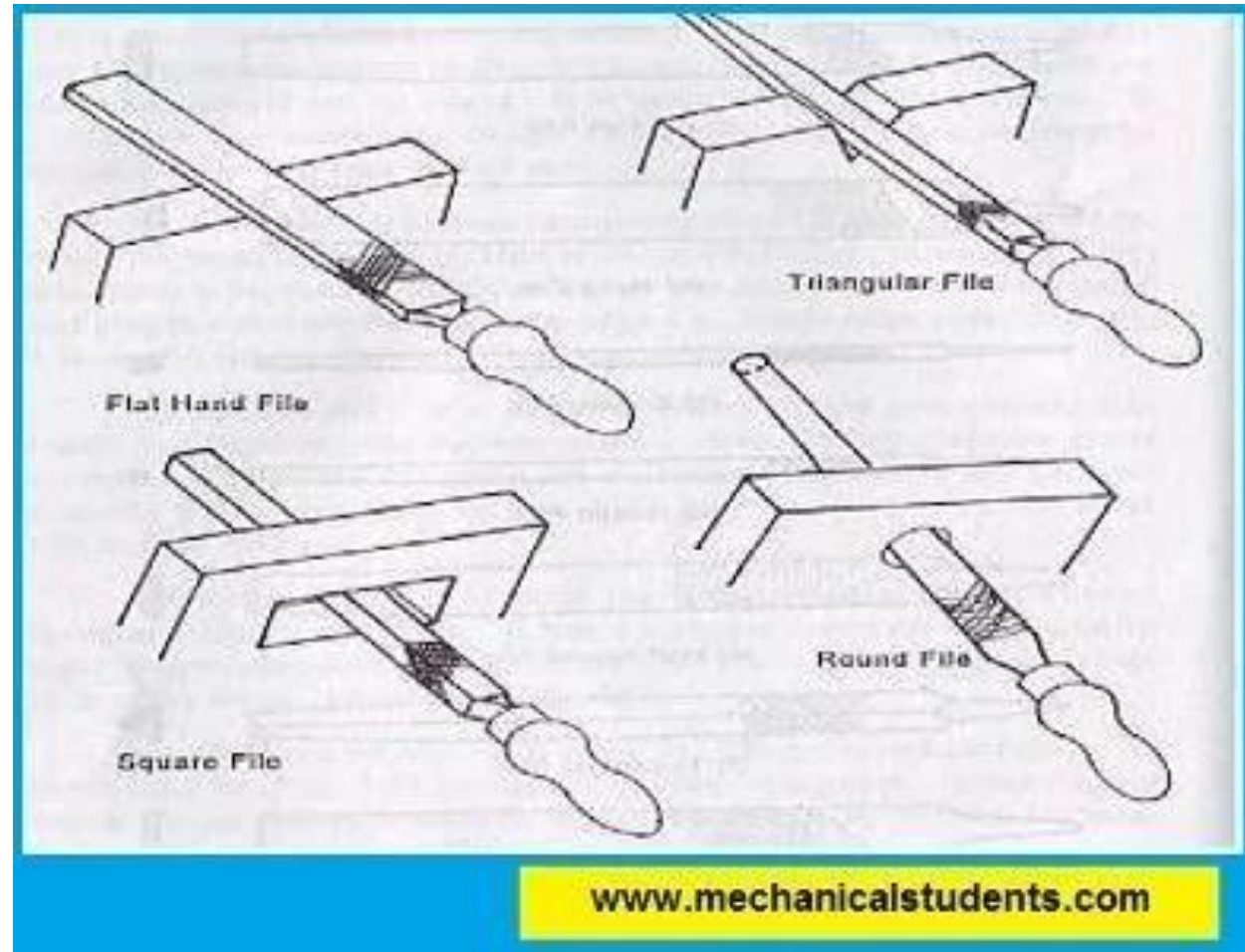


Die with a die handle

Chapman, W.A.J. (1972). Workshop Technology: Part I. and Part 2. Routledge.
<https://mechanicalnotes.com/fitting-workshop/>
https://en.wikipedia.org/wiki/Tap_and_die
<https://www.docsity.com/en/workshop-practice-tools/2423906/>

FINISHING TOOLS

- Flat file
- Square file
- Triangular file
- Round file
- Half-round file
- Swiss or Needle file



Different types of files

STRIKING TOOLS

- Cross-peen hammer
- Claw hammer
- Mallet



Cross-peen hammer



Claw hammer



Wooden mallet

We are at the end of this section

- Now you should be able to
 - LO-6: Use of basic hand tools and machine tools in engineering workshop for machining processes and joining metals
 - **Fitting work**



END